AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 5 · WORKBOOK

Pictures Become Numbers

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected image-representation record.

Decode a grid, restore missing shape information, test a count feature, diagnose a convention mismatch and design a new pair.

Keep one habit: Keep the values, their positions and the code that gives them meaning.

MAIN LESSON · SECTION 1

Recall · Rules and information paths

> Retrieve both earlier ideas before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Rules or Learned Patterns?. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from **Input**, **Process**, **Output**. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

> **My remembered idea, example and correction**

▼ **Earlier wording, alternative practice and picture transcripts**

Rules or Learned Patterns?

Input, Process, Output

MAIN LESSON · SECTION 12

W1 · Decode and read back

> Use the exact dimensions and colour convention.

Your evidence record

Using 1 for black and 0 for white, draw width 3, height 2 with rows 1,0,1 and 0,1,0. Read it back into numbers and count black **pixel**s.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Preserve the missing field

> Show two shapes from the same ordered values.

Your evidence record

A message contains 1,0,0,1,1,0 but gives no dimensions. Show a one-row arrangement and a two-row arrangement using the same values. Name the extra information that would distinguish them.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Test a count feature

> Construct another arrangement with the same count as the cross.

The reference grid

Reference cross: width 3, height 3; rows 0,1,0 / 1,1,1 / 0,1,0. Here 1 is black and 0 white. Use these facts even if you are working without the student lesson open.

Create and explain your counterexample

- A classifier receives only the number of black **pixel**s. Can it always distinguish the chapter's cross from every other three by three pattern? Construct a different pattern with the same count and explain.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Find the earlier input problem

> Check the encoding and decoding convention before changing a classifier.

Your evidence record

A program receives the correct **grid** but displays black wherever the code says 0 and white wherever it says 1. A student proposes collecting more classifier examples.

Identify the earlier mismatch and the first check to perform.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Design a new equal-count pair

> Keep one first attempt before hints or feedback.

Read and explain independently

For W4 use 1 for black and 0 for white, top row first and left to right within each row. Width 4 and height 2 means two rows of four values per image.

Create two different images of width 4 and height 2, each with exactly three black **pixel**s. Write both complete row **encoding**s. Explain which fact their shared count retains and which difference it discards.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ Earlier wording, alternative practice and picture transcripts

Independent case: Design a new counterexample

MAIN LESSON · SECTION 9

Project · Preserve the needed spatial clues

> Record which visual measurements support the sorter's task.

Your project decision

- Choose which visible item features the Classroom Sorter may represent as image measurements and name one feature that lighting could distort.

- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the original attempt and a changed-case return check.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Point to a reconstruction, counterexample and limit.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Missing-field practice copy**

REFERENCE · SECTION 2

Reference · Missing-field practice copy

> This is the same W2 attempt. Use one response record.

Use this when helpful

Use W2: Preserve the missing field for Images as numerical **grid**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Is 1 always black?

REFERENCE · SECTION 3

Extra practice · Is 1 always black?

> Use the declared convention as evidence.

Use this when helpful

Use the worked case for Images as numerical **grid**s. Proposed explanation: “The number one always means black”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · The reading-order target

REFERENCE · SECTION 5

Reference · The reading-order target

> Use the target to explain why the same values can occupy different positions.

Use this when helpful

Encode a 2×3 monochrome image with top row black–white–black and bottom row white–black–white, using 1 for black. A partner reads columns first. Explain why the image changes although the six values are identical.

▼ **Extra practice · Change one square**

REFERENCE · SECTION 6

Extra practice · Change one square

> Try before checking the worked explanation.

Use this when helpful

A 2 by 2 image has rows [0,1] and [1,0]. Zero means white and one means black.

Draw it, then change the top-left value to 1.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · Reverse the colour convention**

REFERENCE · SECTION 7

Extra practice · Reverse the colour convention

> Read the new code carefully; this is additional practice.

Use this when helpful

Use 0 for dark and 1 for light. Draw a two-by-two image with rows [1,0] and [0,1]. Then change only the bottom-left **pixel** to light. Which number changes?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 5 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before drawing or transforming.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Reading-order response space

REFERENCE · SECTION 11

Reference · Reading-order response space

> The teaching case is practice rather than fresh assessment.

Use this when helpful

Encode a 2×3 monochrome image with top row black–white–black and bottom row white–black–white, using 1 for black. A partner reads columns first. Explain why the image changes although the six values are identical.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · New counterexample response copy

REFERENCE · SECTION 15

Reference · New counterexample response copy

> This duplicates W4. Reuse its first-attempt record.

Use this when helpful

Create two different images of width 4 and height 2, each with exactly three black **pixel**s. Write both complete row **encoding**s. Explain which fact their shared count retains and which difference it discards.

> **My reasoning and evidence**

▼ Extra practice · Read the badge picture

REFERENCE · SECTION 18

Extra practice · Read the badge picture

> Use the worked image and record any support.

Use this when helpful

Two fictional 2-by-3 **pixel** badges contain the same number of dark squares.

Dark=1, light=0; rows are read top to bottom, left to right.

Badge A rows: 1,0,1 / 0,1,0. Badge B: 1,1,1 / 0,0,0.

AI ACADEMY / AI-1/5

Pixel positions: two badges, same dark count

	Left	Middle	Right
A top	1	0	1
A bottom	0	1	0
B top	1	1	1
B bottom	0	0	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/5

Pictures Become Numbers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Pixel convention

Each image has two rows and three columns; 1 means dark.



2. Badge A

Top 101, bottom 010: three dark pixels.



3. Badge B

Top 111, bottom 000: also three dark pixels.



4. Compare arrangements

The totals agree, but middle-top and middle-bottom positions differ.

CHANGE ONE THING

Change A's bottom-left pixel from 0 to 1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Find a position where the original badges differ, then read the changed A back as a picture.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: If the sorter uses images, retain dimensions and arrangement rather than assuming a dark-**pixel** count describes shape.

▼ **Reference · Sorter portfolio record**

REFERENCE · SECTION 19

Reference · Sorter portfolio record

> Extend your main project record if needed.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

If the sorter uses images, retain dimensions and arrangement rather than assuming a dark-**pixel** count describes shape.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: image pictures and feedback

REFERENCE · SECTION 20

Extra practice: image pictures and feedback

> Attempt each question before opening its explanation. Keep this practice separate from the reserved new case.

First supported question

- Two fictional 2-by-3 **pixel** badges contain the same number of dark squares.
- Which labelled step matches this detail? The totals agree, but middle-top and middle-bottom positions differ.
- 1. Compare arrangements
- 2. Badge B
- 3. **Pixel** convention
- 4. Badge A

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Compare arrangements The totals agree, but middle-top and middle-bottom positions differ. A **grid** preserves where each brightness belongs.

Adding the six values preserves only the dark count, which is 3 in both images. The total therefore cannot distinguish these badges. Dimensions and reading order belong to the representation, not to an optional caption.

Changed-pixel question

- Changed-case practice: Change A's bottom-left **pixel** from 0 to 1. Which conclusion is supported by the supplied facts?
- 1. A now has four dark **pixel**s; its row **encoding** is 101 / 110.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed pixel**

After your attempt, compare with the supported explanation: A now has four dark **pixel**s; its row **encoding** is 101 / 110. Exactly one light **pixel** becomes dark; the other five positions remain unchanged.

Repair and save

Review Rules or Learned Patterns? (ai-1/4). Goal: Execute a stated rule including its boundary. Short practice: Show which stage used labels and which stage used only the new measurement..

Return to this week and explain what you changed.

Review **Input**, **Process**, **Output** (ai-1/2). Goal: Trace information through a system without confusing information with a physical action. Short practice: Use the richer record to find blue pencils in Box B.

Explain why total alone was insufficient.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternative practice and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose a relevant image rather than repeating all six.

01. Pictures Become Numbers: the case

AI ACADEMY / AI-1/5 · 01 · LOOK AT THE FACTS

Pictures Become Numbers: the case

The situation

Two fictional 2-by-3 pixel badges contain the same number of dark squares.

Supplied evidence

Dark=1, light=0; rows are read top to bottom, left to right.
Badge A rows: 1,0,1 / 0,1,0. Badge B: 1,1,1 / 0,0,0.

By the end, I can encode pixel positions and show why a count cannot reconstruct a picture.

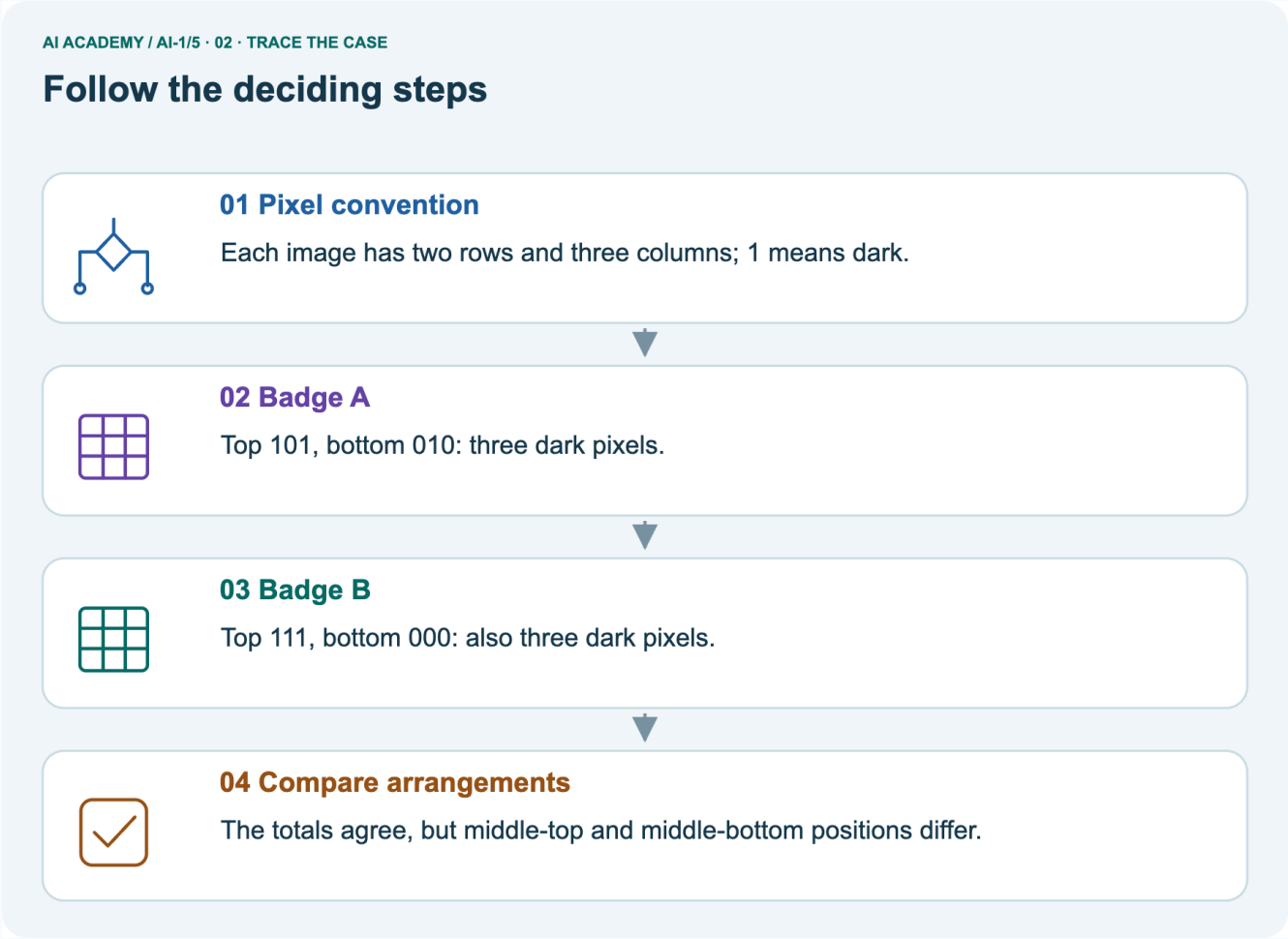
Follow the labelled picture with the matching explanation. Open the vector diagram

Two fictional 2-by-3 **pixel** badges contain the same number of dark squares. Supplied facts: Dark=1, light=0; rows are read top to bottom, left to right. Badge A rows: 1,0,1 / 0,1,0. Badge B: 1,1,1 / 0,0,0.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



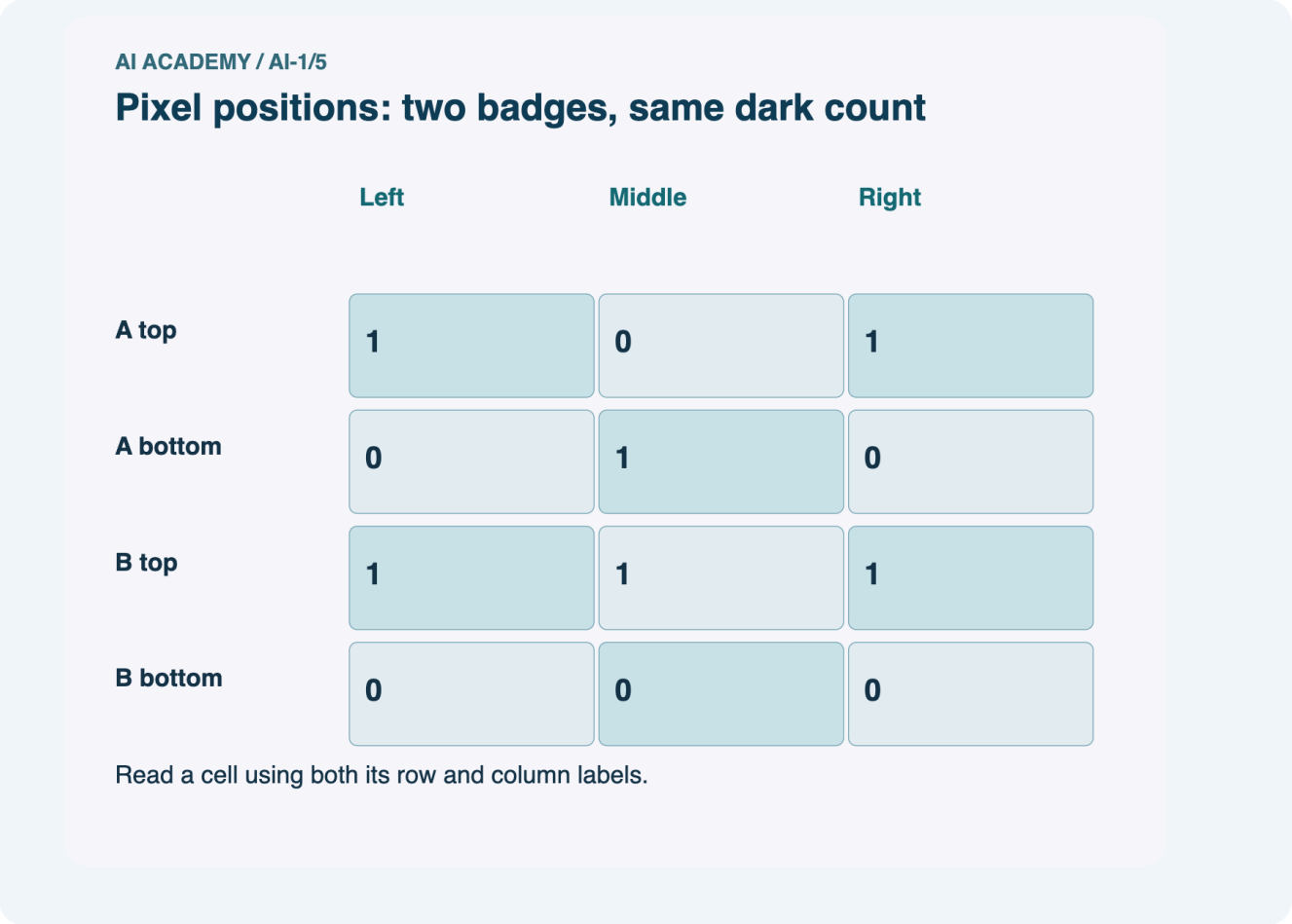
Follow the labelled picture with the matching explanation. Open the vector diagram

Pixel convention: Each image has two rows and three columns; 1 means dark. → Badge A: Top 101, bottom 010: three dark **pixel**s. → Badge B: Top 111, bottom 000: also three dark **pixel**s. → Compare arrangements: The totals agree, but middle-top and middle-bottom positions differ.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Pixel positions: two badges, same dark count



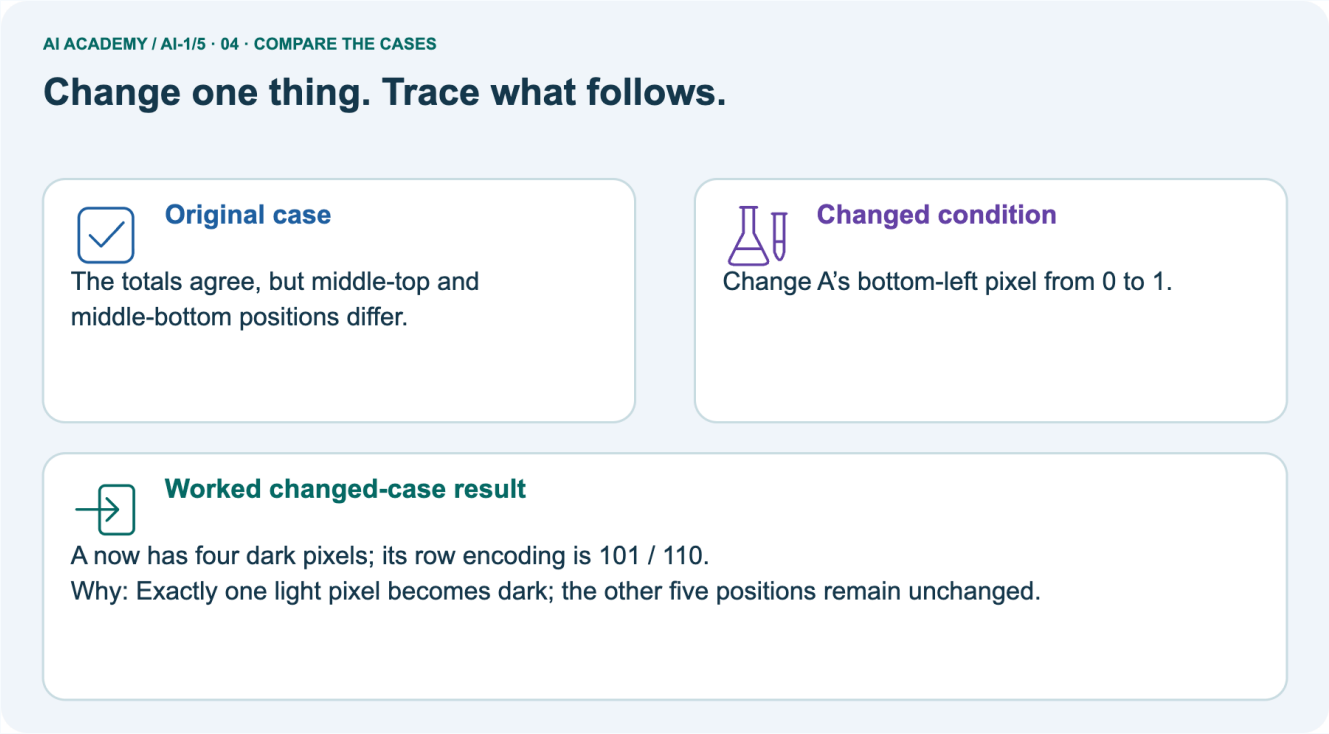
Follow the labelled picture with the matching explanation. Open the vector diagram

Pixel positions: two badges, same dark count. A top: Left = 1, Middle = 0, Right = 1. A bottom: Left = 0, Middle = 1, Right = 0. B top: Left = 1, Middle = 1, Right = 1. B bottom: Left = 0, Middle = 0, Right = 0.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

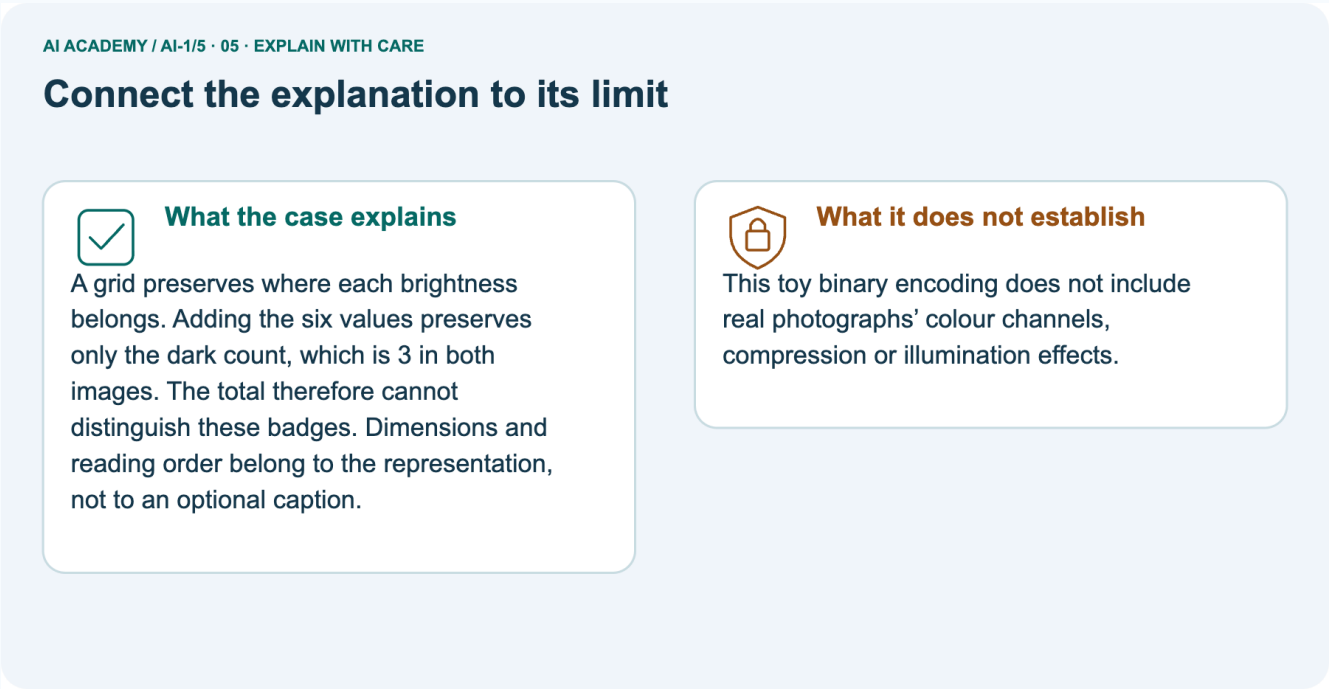
Original case: The totals agree, but middle-top and middle-bottom positions differ. Changed condition: Change A's bottom-left **pixel** from 0 to 1. Expected result: A now has four dark **pixel**s; its row **encoding** is 101 / 110.

Reason: Exactly one light **pixel** becomes dark; the other five positions remain unchanged.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

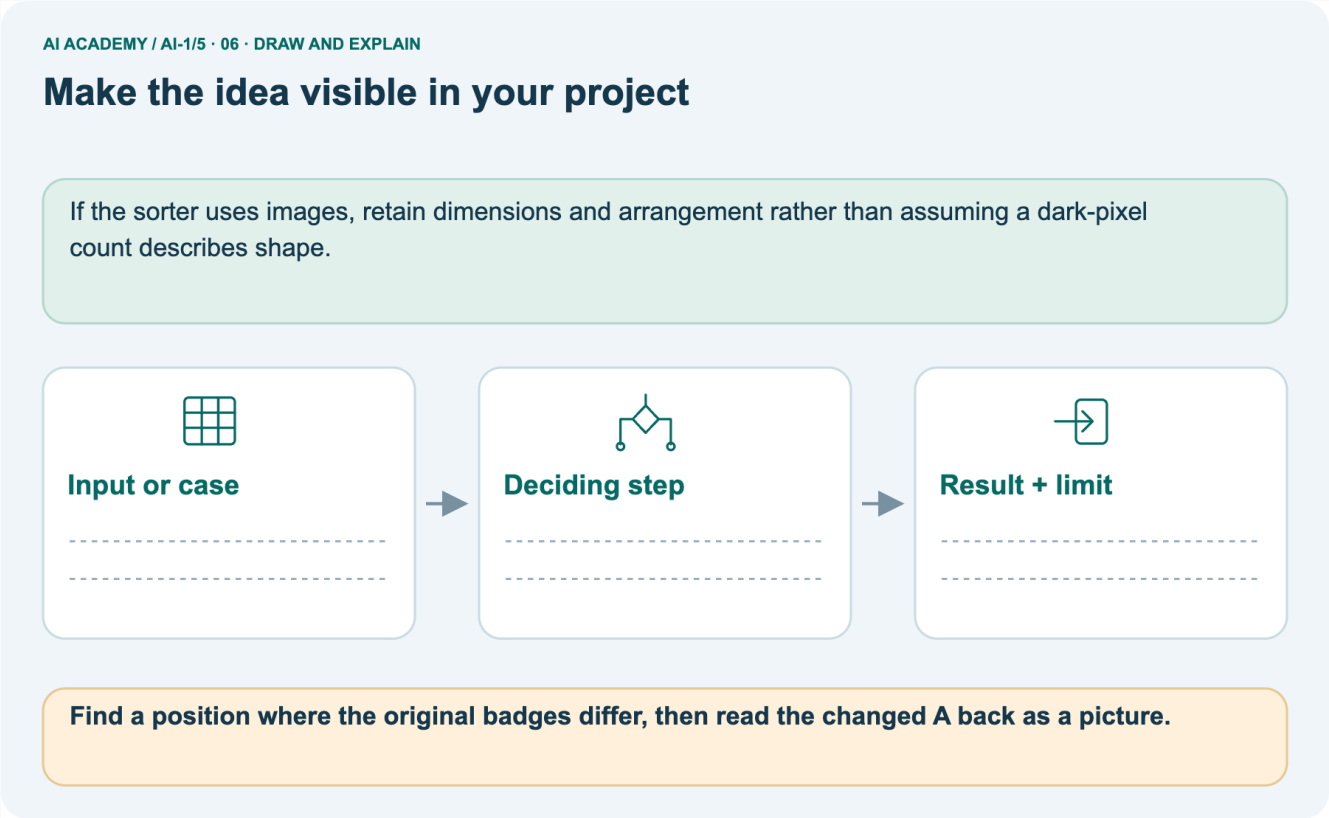
A **grid** preserves where each brightness belongs. Adding the six values preserves only the dark count, which is 3 in both images. The total therefore cannot distinguish these badges.

Dimensions and reading order belong to the representation, not to an optional caption. Limit: This toy binary **encoding** does not include real photographs' colour **channel**s, compression or illumination effects.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

If the sorter uses images, retain dimensions and arrangement rather than assuming a dark-**pixel** count describes shape. Draw your **input**, deciding step, result and limit.

Supported question: Find a position where the original badges differ, then read the changed A back as a picture.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternative practice and picture transcripts

01. Pictures Become Numbers: the case

02. Follow the deciding steps

03. Pixel positions: two badges, same dark count

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.